Land cover / land use classifier: test on 500 records

Run 20261012-151353 · 500 random records (seed 42) of 25,129 · both taxonomies, depth 3 · mistral-small-latest, 5 records/call, 8 parallel · 12 Oct 2026

11 min 29 s
total time (0.73 records/s)

$0.183
total cost ($0.37 per 1,000)

462
API calls

936k
tokens (about 1,872 per record)

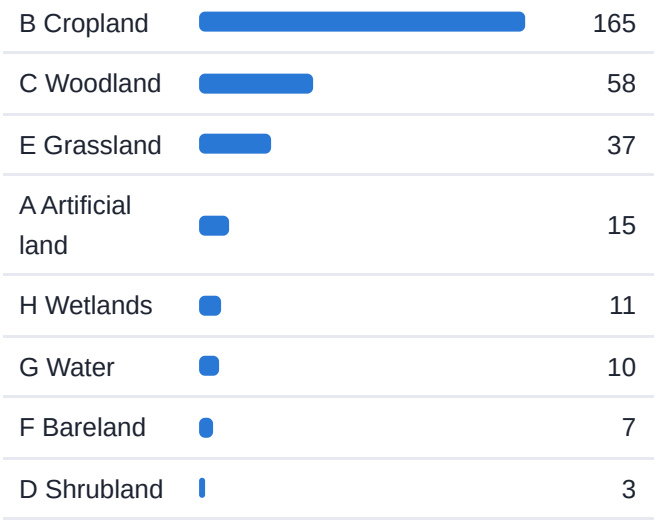
Time and cost

METRIC	VALUE
Wall time	689 s (1.38 s per record)
Input / output tokens	841,323 / 94,913
Cost per record	$0.00037 (prices: $0.15 / $0.60 per million input / output tokens)
Extrapolated to all 25,129 records	about 9.6 h, about $9.20

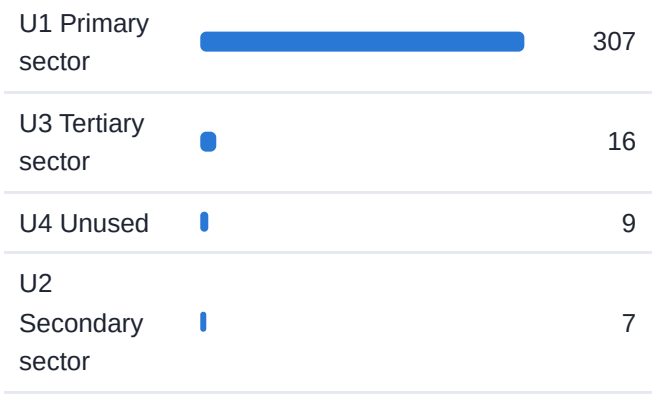
Label coverage and confidence (default thresholds)

	LAND COVER	LAND USE
Records with at least 1 label	284 (57%)	337 (67%)
Records reaching level 2 / level 3	37% / 20%	52% / 42%
Labels per record (deepest)	0.71	0.71
Mean level-1 confidence	0.94	0.92
Level-1 labels with confidence ≥ 0.8	100.0%	99.7%

Land cover: level 1 classes



Land use: level 1 classes



Accuracy 1: independent check by Claude (60 records)

The records table has no human labels. Claude labelled 60 of the 500 records (random, seed 7) at level 1 **before** seeing Mistral's answers, from the title and the first 650 characters of the abstract (Mistral saw the full text). Mistral's labels use the default thresholds. Details: `reports/claude_verification_60.csv`.

LEVEL 1	EXACT MATCH	PRECISION	RECALL	F1
Land cover	47/60 (78%)	0.82	0.79	**0.81**
Land use	40/60 (67%)	0.54	0.73	**0.62**

Land cover is good: most of the 13 differences are a second class Mistral missed (wetland, shrubland, bare soil); in 3 cases it added one Claude did not.
Land use has a clear problem: Mistral puts U1 (Primary sector) on soil-science records that are not about farming, forestry, fishing or mining (biochar, an E. coli lab column, soil enzymes, a global salinity map). 14 of the 20 differences are this, and U1 is 307 of 337 labelled land-use records in the whole run. **Suggested fix:** tighten the U1 hint in the Labels tab; a changed hint reclassifies the records once (about $0.18 per 500).
Limits: one annotator (some labels are judgment calls), level 1 only, small sample. This is a second opinion from a stronger model, not human ground truth.

Land use: the 20 differences

#	TITLE	CLAUDE	MISTRAL
9	The power of pyrolysis	-	U1
10	Effect Of Elevated Co2 And Temperature On The Oxidative Stress Respons	-	U1
15	International workshop from yield-based to society-based fertilizer re	U1	-
16	Cyanobacteria for restoration of degraded soils	-	U1
20	Physiological And Biochemical Responses Ofquercus Pubescensto Air Warm	-	U1
21	ORCHILEAK (revision 3875)	-	U1
26	Bacteriophages limitedly contribute to the antimicrobial resistome of	U3	-
27	Remediation of water from perfluorooctanoic acid (PFOA) with ozonation	-	U2
29	Nutrient Additions To A Tropical Rain Forest Drive Substantial Soil Ca	U4	U1
37	Biotic and abiotic factors controlling spatial variation of mean carbo	-	U1
39	Biochar Pyrolyzed At Two Temperatures Affects Escherichia Coli Transpo	-	U1
41	Soil scans for parcels in Netherlands (20170327_vermeulen_spieClass_3)	U1	-
42	Different Types Of Sub-Alpine Grassland Respond Similarly To Elevated	-	U1
47	The Invasive Shrub European Buckthorn (Rhamnus Cathartica, L.) Alters	U4	U1
51	Cessation of grazing causes biodiversity loss and homogenization of so	U1,U4	U1
52	Elevated Co2 Alleviates Damage From Potato Virus Y Infection In Tobacc	-	U1
53	From fine to giant: multi-instrument assessment of the dust particle s	-	U1
56	Global Soil Salinity Map	-	U1
58	Farmwell social innovation Greece energy community	U2	U1
59	SUMMARY OF THE PM(T) ONLINE SURVEY RESULTS - MAY-JULY 2022	-	U2

Land use: fixing the "Primary sector" over-labelling

Mistral Small puts U1 (Primary sector) on soil-science records that are not about farming, forestry, fishing or mining. Five setups (the last is an offline estimate) were scored against Claude's level-1 labels on two sets, and their union: the **first 60** records (used to tune the hint and the gate wording, so slightly optimistic) and **40 fresh** records labelled blind afterwards (the fairer test). Gate = one extra cheap call per record asking whether it concerns land that is actually used or unused; records scoring below 0.5 get no land use label.

SETUP	FIRST 60		FRESH 40		ALL 100				COST (500 RECORDS, LAND USE)
	EXACT	F1	EXACT	F1	EXACT	PREC.	RECALL	F1	
Small, original hint, no gate	40/60	**0.62**	22/40	**0.56**	62/100	0.51	0.72	**0.59**	$0.18 (both taxonomies)
Small, new U1 hint, no gate	42/60	**0.64**	25/40	**0.60**	67/100	0.56	0.70	**0.62**	$0.085
Small, new U1 hint + gate (current)	45/60	**0.68**	26/40	**0.59**	71/100	0.62	0.65	**0.64**	$0.099
Medium, new U1 hint, no gate	42/60	**0.67**	28/40	**0.71**	70/100	0.60	0.80	**0.69**	$0.248
Medium + gate (offline estimate, not run)	46/60	**0.68**	28/40	**0.70**	74/100	0.66	0.72	**0.69**	about $0.25 + $0.03

Result: the tighter hint and the gate each help a little. Small goes from F1 0.59 (original) to 0.64 (hint + gate) over all 100 records. **Mistral Medium is better: F1 0.69 over all 100 (0.71 on the fresh 40), mainly through higher recall (0.80 vs 0.65), at about 2.5x the land use cost** ($0.25 vs $0.10 per 500 records; roughly $12 vs $5 for all 25,129 records). Medium plus the gate gives the best exact match (74/100) at the same F1. Small was kept as the default for cost; this table is the evidence if that decision is revisited. Medium still puts U1 on soil-science records, and the gate passed 335 of 500 records and still lets some datasets, maps and project reports through. The first 60 records were also used to tune the hint and gate wording, so the fresh 40 are the fairer test. Run-to-run variation is about ±0.03 F1. The labels are one reviewer's (Claude), not human ground truth. Details: `reports/claude_verification_60.csv`, `reports/claude_verification_fresh40_landuse.csv`.

Accuracy 2: agreement with an earlier run

Agreement with run 20261007-162217 (1 record per call) on the 100 records both runs contain. The reference is not human-verified: this measures consistency, not correctness. Precision, recall and F1 treat the reference labels as truth.

LEVEL	LAND COVER				LAND USE			
	PREC.	RECALL	F1	EXACT	PREC.	RECALL	F1	EXACT
1	88%	85%	0.86	85%	94%	89%	0.92	90%
2	78%	78%	0.78	83%	85%	76%	0.80	80%
3	91%	83%	0.87	94%	82%	71%	0.76	79%